

Talking matrices

And other prerequisites

aNLP M'26

August 8, 2026

IIIT Hyderabad

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recurrence
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optimus prime
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muppets!
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WORD EMBEDDINGS

Word Embeddings

- ▶ Before neural networks, we had deterministic methods. Problem: Each word was treated as an atomic entity and there were no relations between them.
- ▶ To model relations, one of the earlier methods was SVD. Problem: Huge sparse matrices, which made the process difficult
- ▶ Cue Word2Vec. Basic premise: a word is known by the company it keeps

Word2Vec

- ▶ CBOW (Continuous Bag of Words): Predict the target word from the surrounding words
- ▶ SkipGram: Predict the surrounding words from the target word
- ▶ Negative Sampling: Sample positive and negative samples to reduce computation.

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MLP

MLP

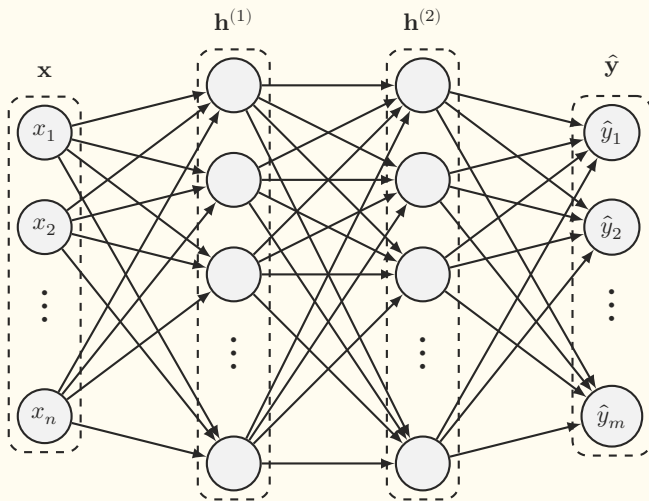


Figure: Multi Layer Perceptron

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RECURRENT NEURAL NETWORKS

Start from the deep end

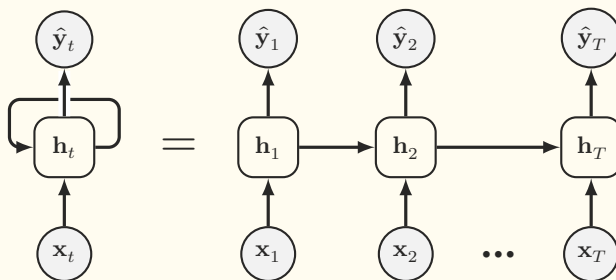


Figure: Classic RNN

What's wrong with an MLP?

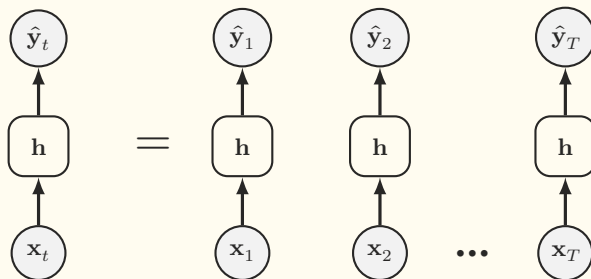


Figure: Classic MLP (Oops)

Lemme check under the hood

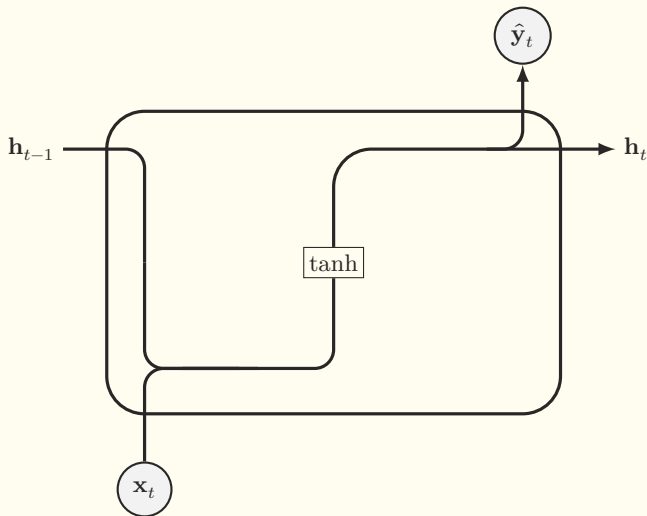


Figure: An RNN cell

Lemme check under the hood

$$\mathbf{h}_t = \tanh([\mathbf{x}_t : \mathbf{h}_{t-1}])$$

Do u rememba da wae?

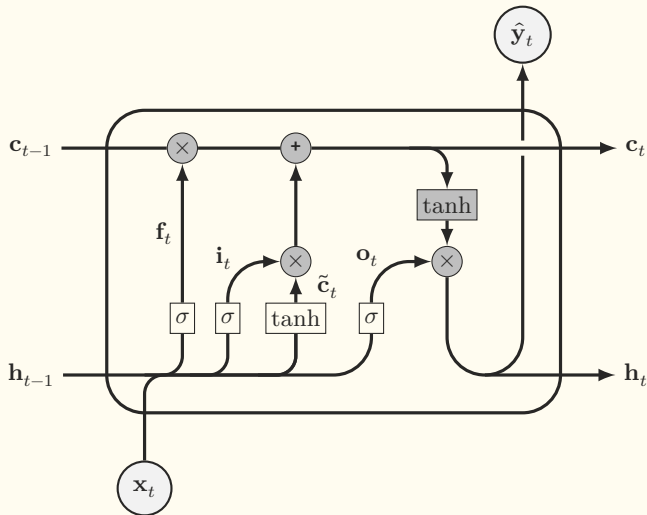


Figure: An LSTM cell

Do u rememba da wae?

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{x}_t : \mathbf{h}_{t-1}])$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{x}_t : \mathbf{h}_{t-1}])$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_{\tilde{\mathbf{c}}}[\mathbf{x}_t : \mathbf{h}_{t-1}])$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{x}_t : \mathbf{h}_{t-1}])$$

$$\mathbf{c}_t = \mathbf{c}_{t-1} \mathbf{f}_t + \mathbf{i}_t \tilde{\mathbf{c}}_t$$

$$\mathbf{h}_t = \mathbf{o}_t \tanh(\mathbf{c}_t)$$

Steal the Moon!

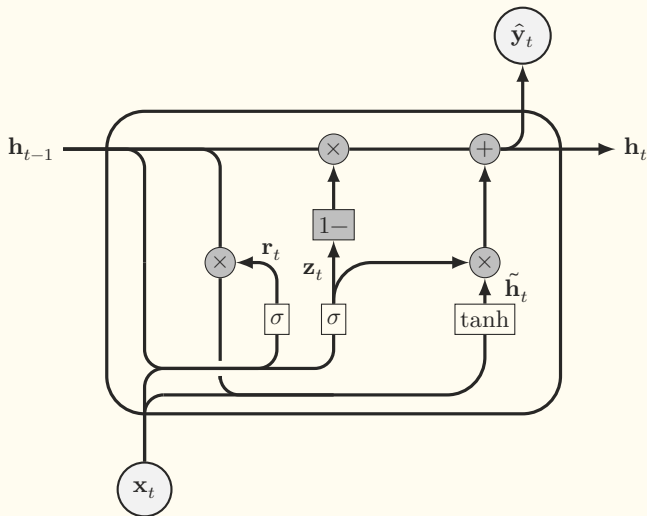


Figure: A GRU cell

Not a one-way street

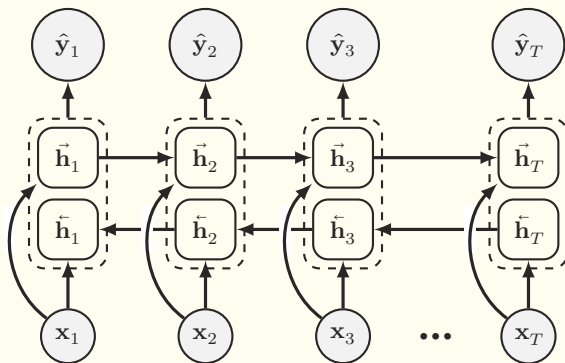


Figure: Backwards!

Bahdanau? Can I be bahda later?

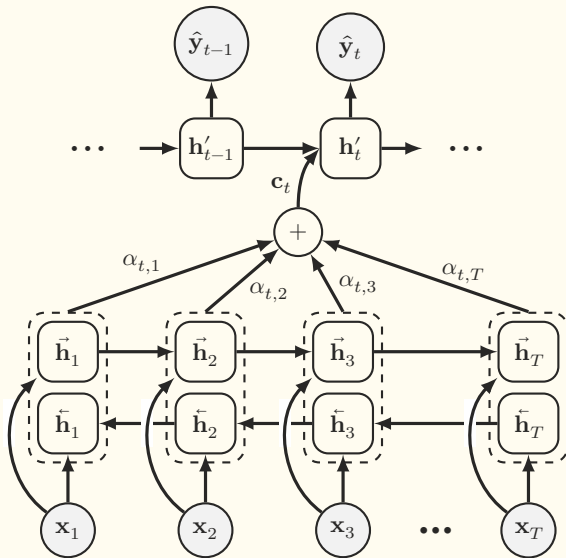


Figure: Attention!

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TRANSFORMERS AND ATTENTION

Pay attention now, y'all

$$\mathbf{q}_i = \mathbf{x}_i \mathbf{W}_Q$$

$$\mathbf{k}_i = \mathbf{x}_i \mathbf{W}_K$$

$$\mathbf{v}_i = \mathbf{x}_i \mathbf{W}_V$$

$$\mathbf{Attention} = \text{softmax} \left(\frac{\mathbf{q} \cdot \mathbf{k}}{\sqrt{d_k}} \right)$$

$$\mathbf{o}_i = \mathbf{Attention} \times \mathbf{v}_i$$

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SOME POPULAR EMBEDDING MODELS

ELMo

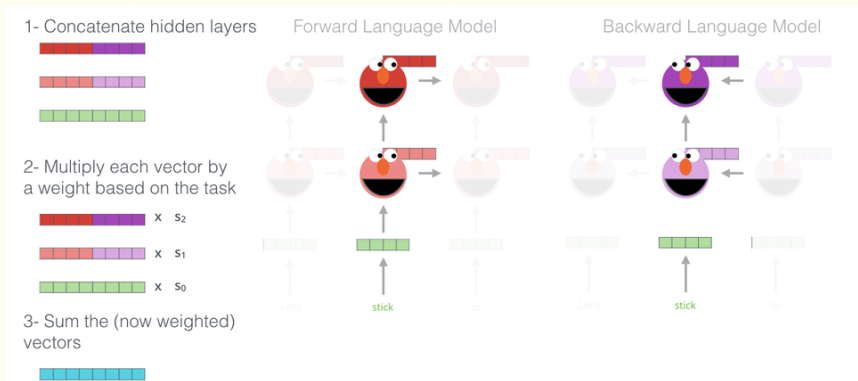


Figure: ELMo is a BiLSTM

Image source: jalamar.github.io/illustrated-bert

BERT?

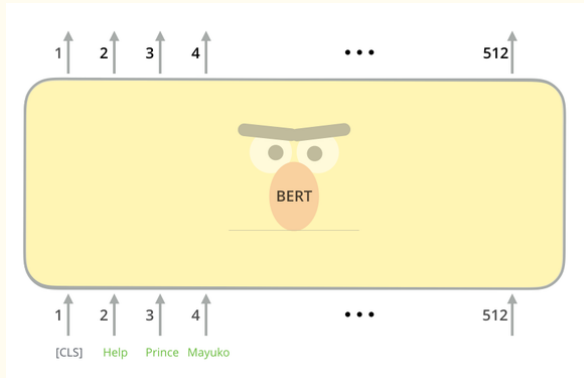


Figure: BERT is a different beast

Image source: jalammar.github.io/illustrated-bert

BERT...

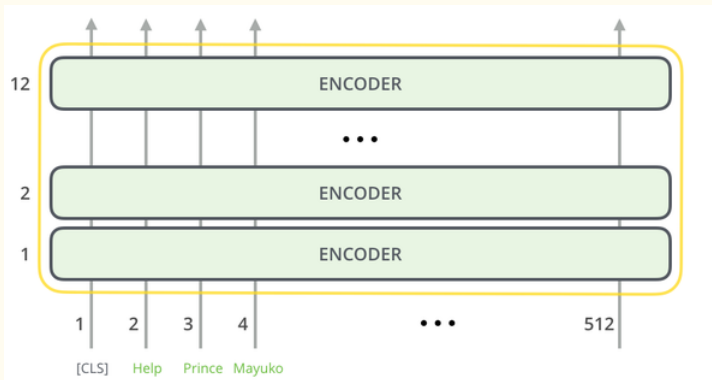


Figure: BERT is a different beast

Image source: jalamar.github.io/illustrated-bert

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